

Bartr

The laundromat on Murray Avenue is worth \$570,768, give or take 22 percent. Today you can buy 20 shares of it.

PITTSBURGH, PRICED TODAY	OUR VALUE
Squirrel Hill Wash and Fold	\$570,768 ±22%
Butler Street Laundromat	\$565,138 ±14%
Steel City Express Car Wash	\$1,534,521 ±37%
Iron City Precision Machine	\$979,576 ±36%
Bloomfield Coin Laundry	\$192,254 ±82%

36.2 million small businesses in America.

SBA Office of Advocacy, 2025. 99.9 percent of all businesses in the country.

80%
**of the ones that try to sell never find a
buyer.**

They shut down. Most of them are profitable.

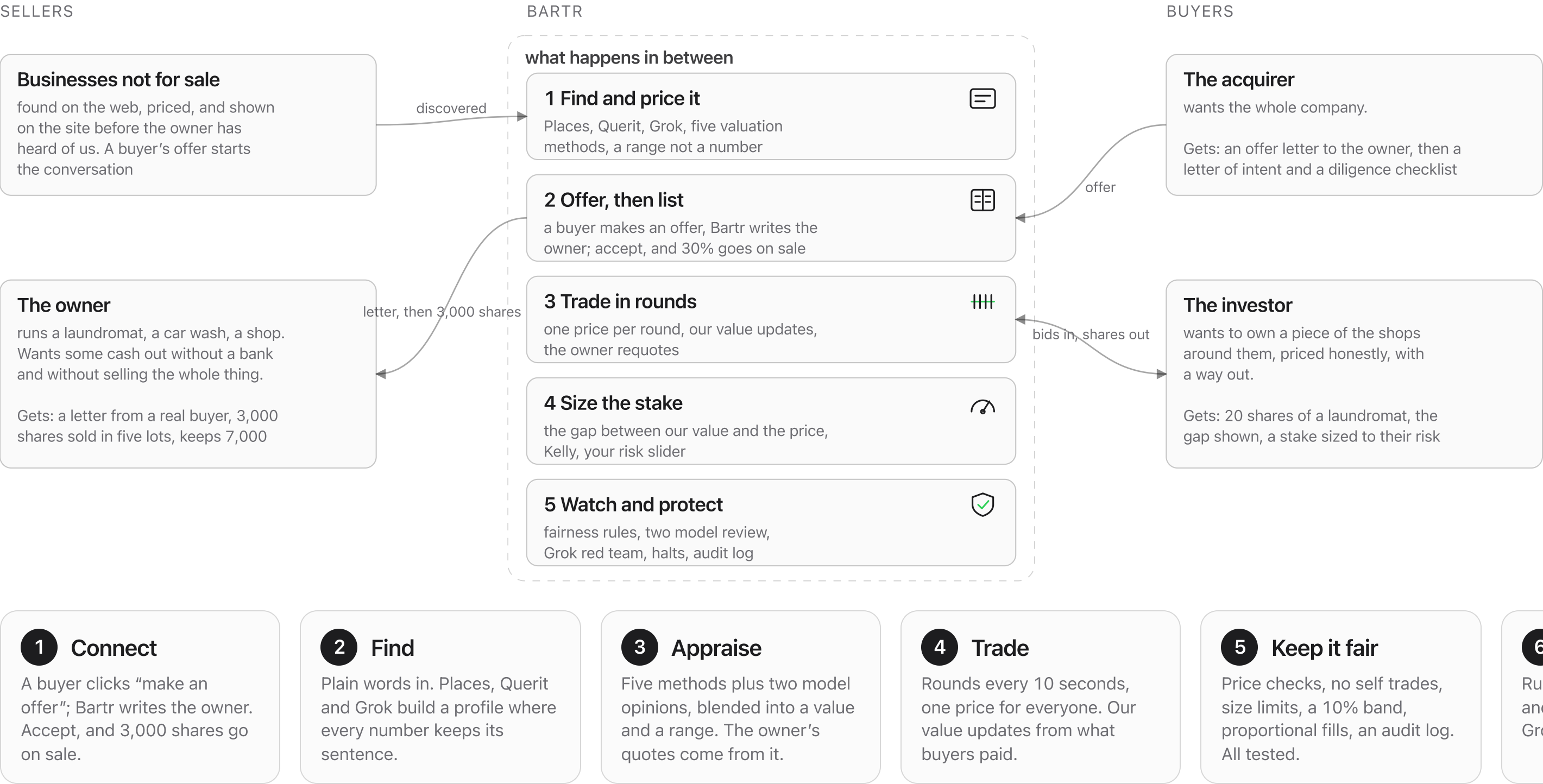
Exit Planning Institute: 20 to 30 percent of businesses that go to market sell. BizBuySell's own close rate is under 7 percent.

Half of the American workforce works for one.

46 percent of private sector jobs, and 9 of every 10 net new jobs last year. SBA Office of Advocacy, 2025.

Who Bartr is for, and what it does

We connect buyers and sellers. Six things, all running.



Discover

Bartr

DiscoverPortfolioSurveillanceAgent

LOGINSTART TRADING

DISCOVER

laundromat in pittsburgh

SEARCH

QUERY

laundromat in pittsburgh

CAT laundromat

NAICS —

STATE PA

STATE

Any

VALUE

Any

SORT

Relevance

3 PRICED · 0 READING

COMPANY	VALUE	BID	ASK	LAST	CONFIDENCE	COMPLETE
Bloomfield Coin Laundry Pittsburgh, PA · laundromat · 4.0 (27)	\$192K	38.86	22.71	25.55		18%
Butler Street Laundromat Pittsburgh, PA · laundromat · 4.4 (118)	\$565K	68.21	58.93	58.91		86%
Squirrel Hill Wash and Fold Pittsburgh, PA · laundromat · 4.6 (214)	\$571K	60.99	57.22	57.22		78%

Bartr

A discovery engine and exchange for the businesses that will never be listed.

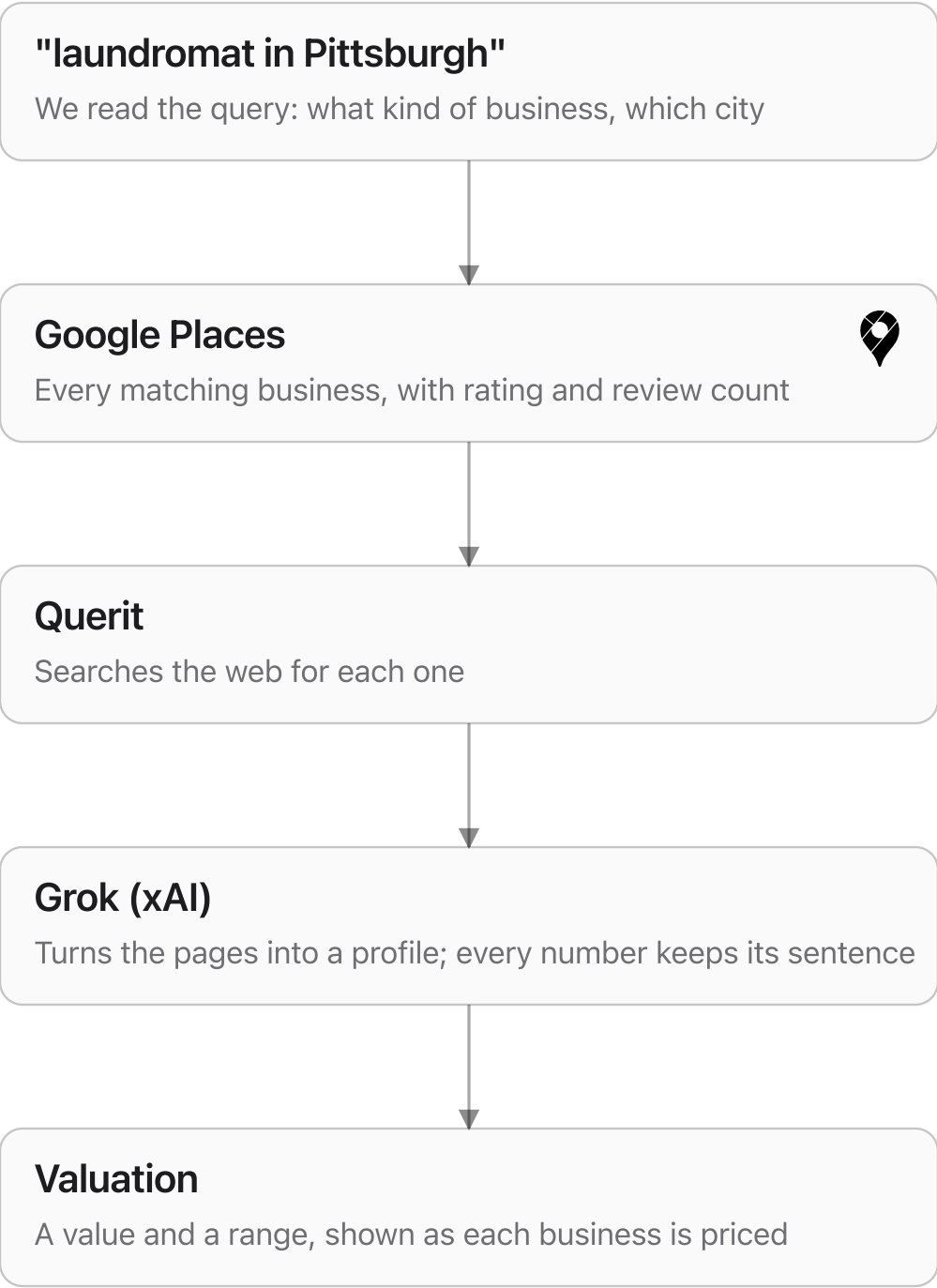
PRODUCT

Discover

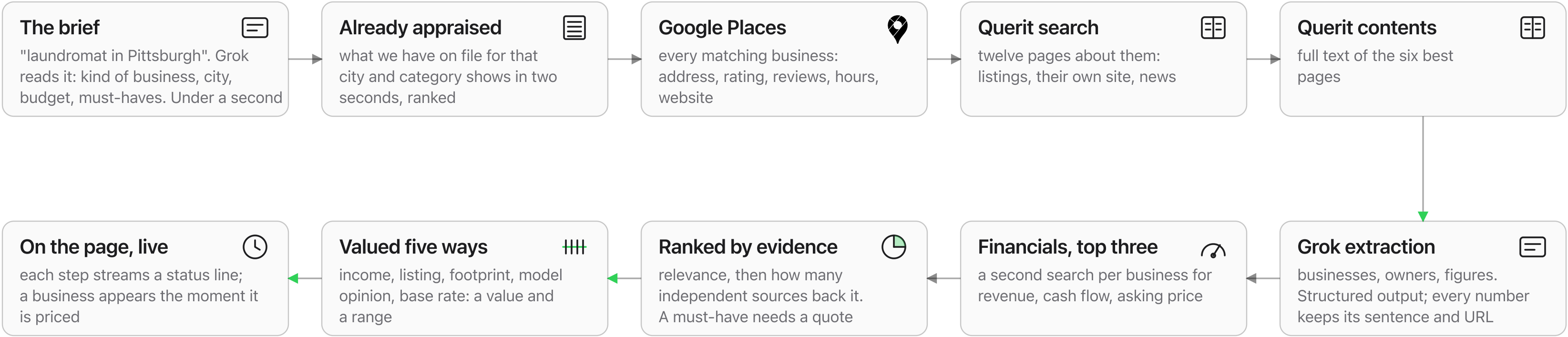
Portfolio

Surveillance

"laundromat in Pittsburgh"



Nothing is typed in by hand. Every business, every number, every source comes from a call we can show you.



Google Places Text Search

Every matching business in the city: name, address, rating, review count, hours, website. This is the list that exists nowhere else in one place.

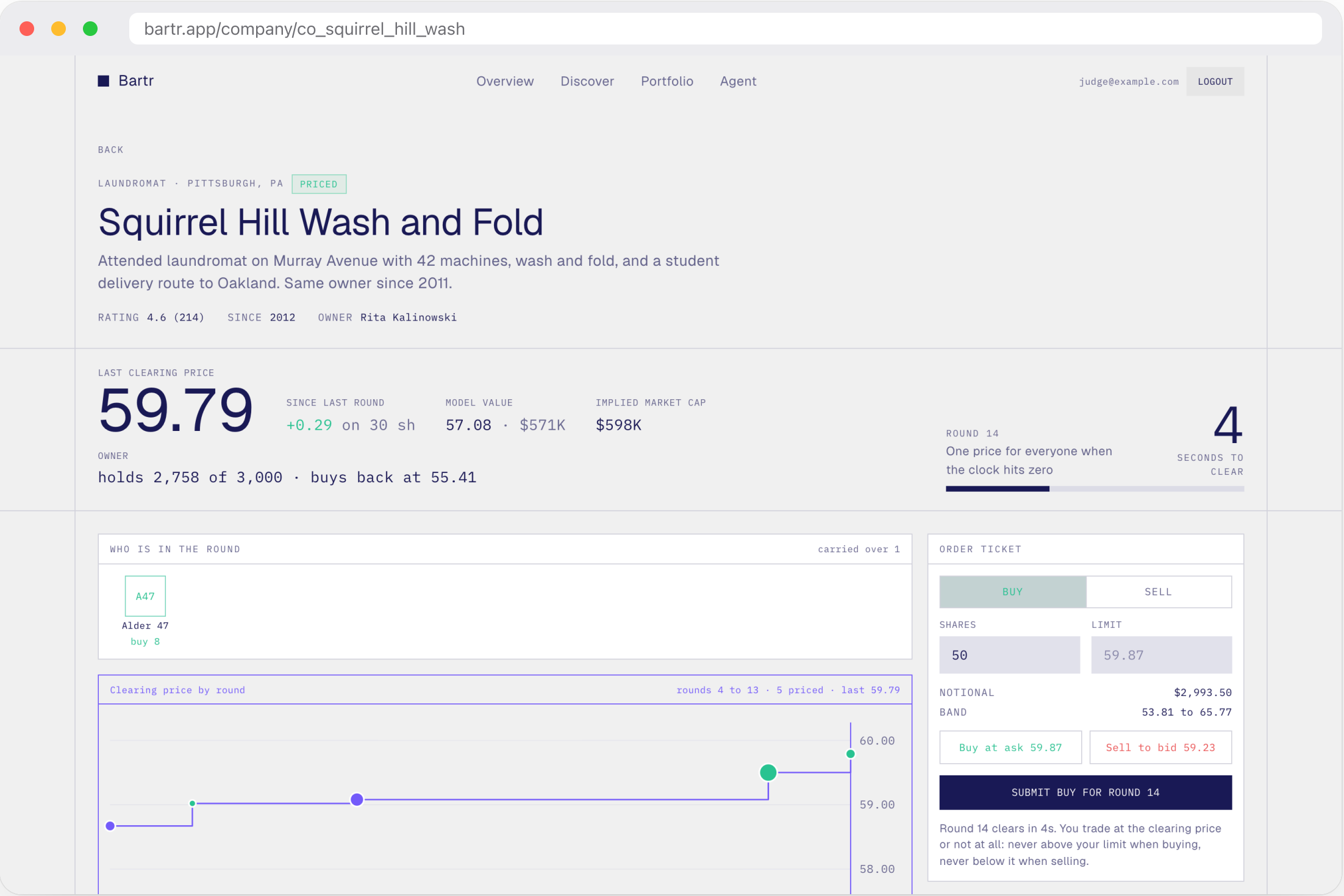
Querit search and contents

Twelve pages per query, then the full text of the best six. Listings, the business’s own site, news, reviews. A targeted second search for the top three looks for revenue, cash flow and an asking price.

Grok, twice

A fast model reads the brief in under a second. A stronger one turns pages into profiles with structured output: every figure keeps the sentence it came from and its URL. Pages are evidence, never instructions.

One company



Squirrel Hill Wash and Fold

\$570,768

Range \$475k to \$686k. We could be about 22 percent off either way.

\$57.08 a share. 10,000 shares.

On the page: the last price, every open order, and the seconds until the next round.

How we price it

and how far off we might be

Five ways to value a business. Each one comes with its own typical miss.

1. What one method says	$\ln V_k = \mu_k \pm \sigma_k$	The income method says Squirrel Hill is worth \$589k, and that method is usually off by about 25 percent. We work in logs so 2x too high and 2x too low count as the same size miss.
2. Combining five answers	$\mu = \frac{\sum_k w_k \mu_k}{\sum_k w_k}, \quad w_k = \frac{1}{\sigma_k^2}$	Each method gets a vote weighted by one over its miss squared. Income (miss 0.25) gets 16 votes, Grok's number (0.35) about 8, the category base rate (0.70) gets 2. The blend lands at \$570,768.
3. How sure is the blend	$\sigma^2 = \underbrace{\frac{1}{\sum_k w_k}}_{\text{each method's own miss}} + \underbrace{\frac{\sum_k w_k (\mu_k - \mu)^2}{\sum_k w_k}}_{\text{how far apart they land}}$	Two things go wrong: every method is a bit off, and the methods disagree. We add both. Squirrel Hill's three methods land between \$469k and \$589k, close together, so the total miss is 0.22.
4. What we show	$[V_{20}, V_{80}] = e^{\mu \mp 0.84 \sigma}, \quad \sigma \geq 0.12$	A range, never a bare number. 0.84 sigma is where the 20th and 80th percentiles sit on a bell curve: \$570,768 times 0.83 is \$475k, times 1.20 is \$686k. We never claim to be surer than 12 percent.

Where each miss comes from

Today's numbers come from BizBuySell's 2026 tables. We also have a check that takes real listings, hides the asking price, runs each method and records the miss. Once it has run on enough listings, measured misses replace this table.

Listing price	it is actually for sale	±15%
Income	owner earnings x category multiple	±25%
Grok's number	wider when K2 disagrees	±35%+
Proxy	revenue guessed from staff, machines, reviews	±60%
Base rate	median for the category in that state	±70%

Bloomfield Coin Laundry has only a Google pin, so two weak methods run: miss 0.82, range \$96k to \$384k. The page shows the wide range instead of hiding it.

The owner is the seller. There is no market maker.

The owner's asks. 3,000 shares (30 percent) in five lots of 600, at five prices just above our value. For Squirrel Hill, value \$57.08: lots at \$58.67, \$60.76, \$63.03, \$65.59 and \$68.62. A shakier valuation spreads those prices wider.

The owner's buyback. The owner also stands ready to buy up to 1,000 shares back at \$47.48, about 17 percent under our value. That bid is in the book from the first minute.

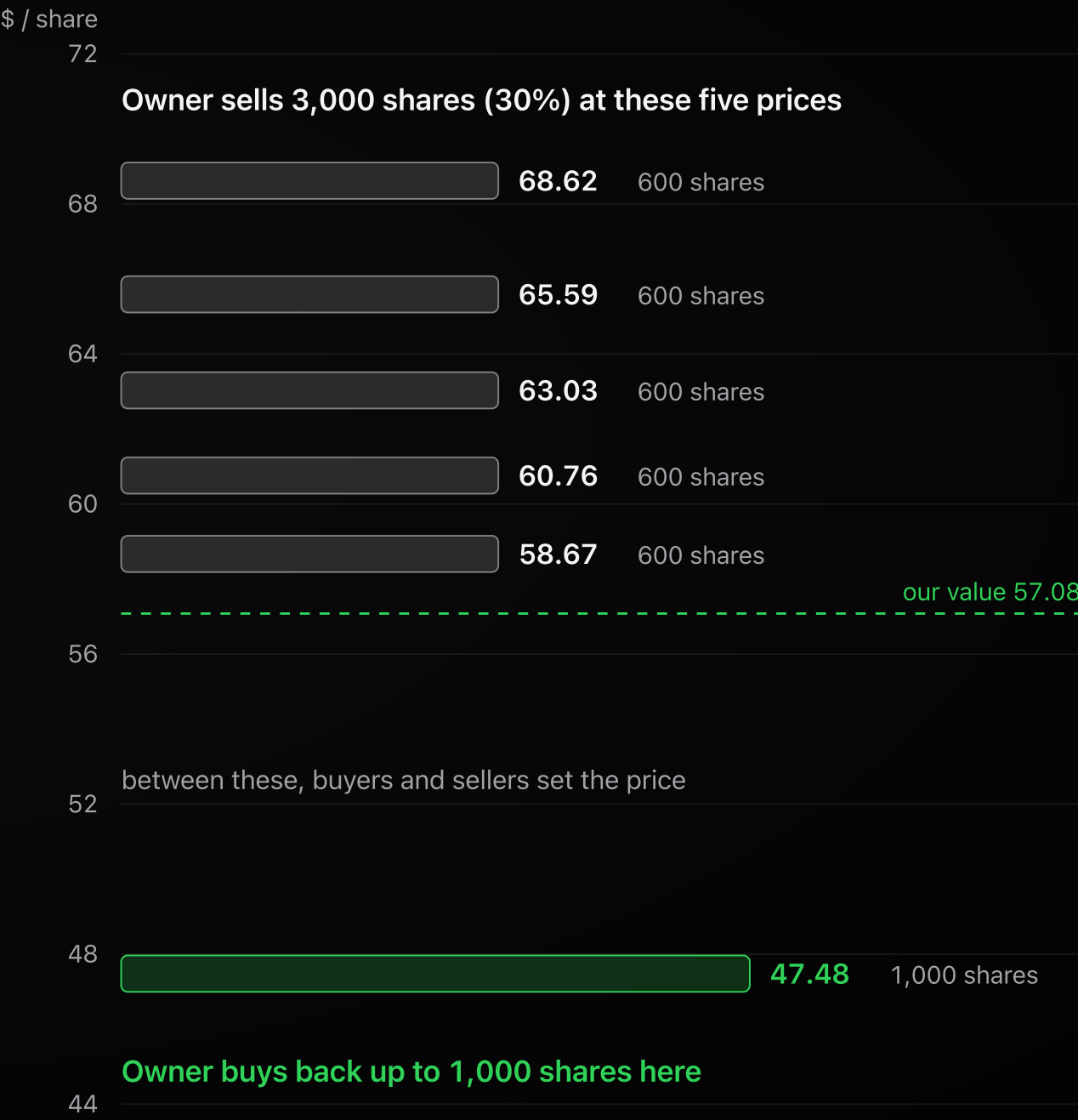
One price per round. Every 10 to 60 seconds, faster when the book is busy, all orders clear at once at a single price. Everyone in that round pays it. Between rounds our value updates from what buyers paid, and the owner's asks and buyback move with it. Bartr holds no shares and takes no side.

Which price does a round clear at?

$$p^* = \arg \max_p \min \left\{ \underbrace{D(p)}_{\text{shares buyers want at } p}, \underbrace{S(p)}_{\text{shares offered at } p} \right\}$$

Try every price. At each one, count how many shares buyers would take and how many sellers would give. Clear where the smaller of the two is largest. In round 1 of the demo, buyers wanted 305 shares at \$58.67 or above and the owner offered 600 there, so 305 traded at \$58.67, and Jonas, who bid \$64.20, paid \$58.67 too.

One price per round. The owner is on the other side.



Live, two rounds

Squirrel Hill Wash and Fold · 10,000 shares · our opening value \$57.08

RK

Rita K.
owner
sold 1,158.85 of 3,000

sell 100 @ 60.20

J

Jonas
Bloomfield
holds 288.85, sold 71.15

buy 150 @ 60.90

P

Priya
Oakland
holds 330

buy 200 @ 62.00

A

Ana
phone, in the room
holds 200

buy 120 @ 63.50

L

Leo
Shadyside
holds 190

buy 60 @ 54.80

T

Tom
Lawrenceville

buy 55 @ 54.60

K

Ken
Greenfield

buy 150 @ 61.50

S

Sam
phone, in the room
holds 150

buy 25 @ 49.10

N

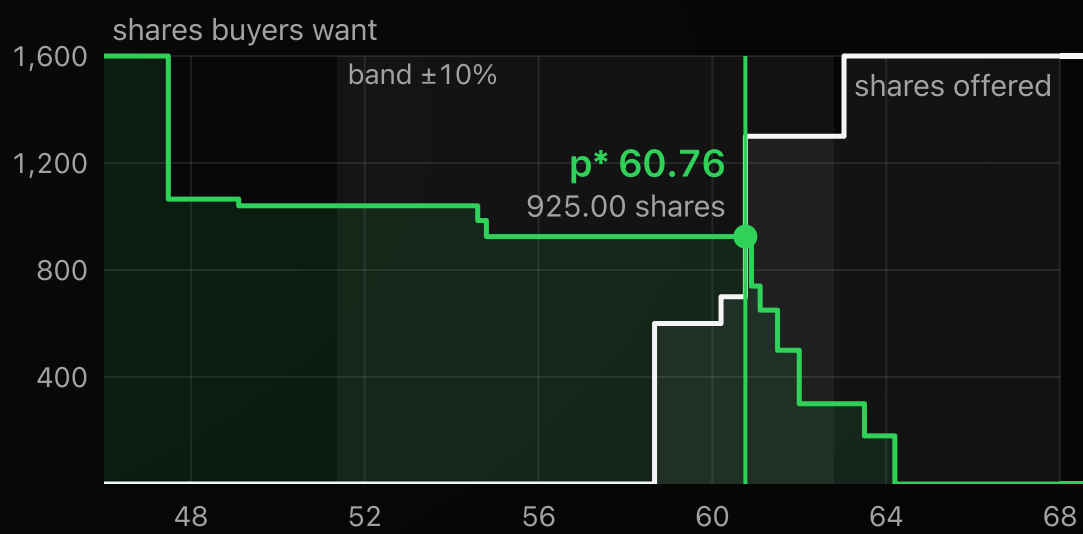
Nina
South Side

Bids		qty · limit	Asks		qty · limit
60.00	54.80	Tom	28.85	60.20	Jonas
55.00	54.60	Ken	173.08	61.35	owner
25.00	49.10	Nina	600.00	62.58	owner
1,000.00	53.65	owner floor	600.00	63.95	owner
			600.00	65.54	owner

After 2 rounds

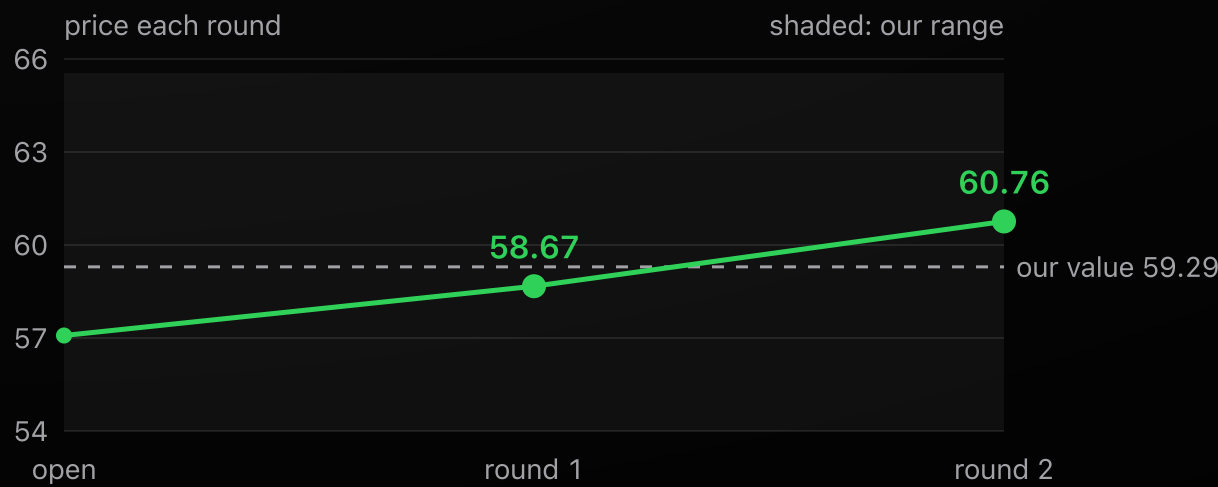
60.76

everyone paid this

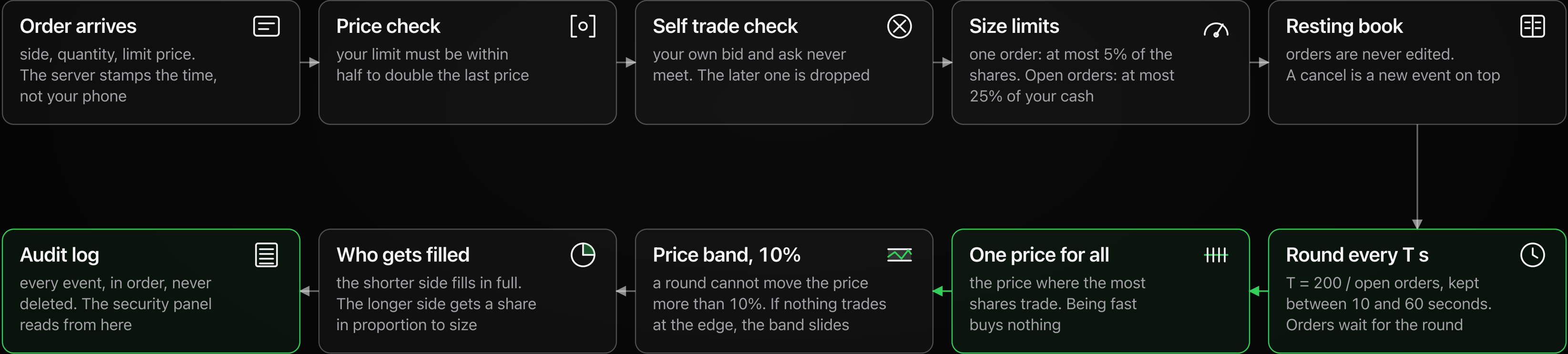


Round 2: 925.00 shares at 60.76. Jonas, Priya, Leo, Ana, Sam all paid 60.76, whatever they bid. 853.85 came from the owner. Jonas sold 71.15 of those to them, at the same price.

Buyers paid more than our 57.08, so our value moves to 59.29. The owner's asks and buyback bid move with it.



One price. No head start.



Nothing matches between rounds, so a faster connection gets you nothing. Two rounds in a row at the band edge halt the market for one round. The owner can only change quotes between rounds, never bids above its buyback price and never asks below its lowest ask, so the account holding 30 percent of the shares cannot chase the price. Every order carries a server timestamp and lands in the audit log.

Every suggestion shows the gap between our value and the market price. A slider decides how hard to lean on it.

The gap. Our value next to the last traded price. We only suggest businesses trading below our value; the gap is what you stand to gain when the price catches up.

How much to buy. The Kelly formula turns the gap and our uncertainty into a share of your cash. A big gap with a sure value gets more; a big gap with a shaky value gets less. No single business ever gets more than 20 percent.

Your risk level. A slider from 0.25 to 1.0 scales that share. Default is half, because our value is an estimate. Move it and the gain and the possible loss move with it.

How much of my cash should go in?

$$\text{gap} = \ln \frac{\text{our value}}{\text{price}}, \quad \text{share of cash} = \text{slider} \times \frac{\text{gap}}{\sigma^2}, \quad \text{capped at 20\%}$$

Iron City: value \$97.96, price \$94.50, gap 3.6 percent, miss 0.36. Slider at 0.5: 0.5 x 0.036 / 0.13 = 14 percent of your cash.

Last traded price

\$94.50

Our value

\$97.96 miss 0.36

Gap, what you stand to gain

+3.6%

Your risk level

0.50

0.25 cautious

1.0 full Kelly

Suggested stake

\$13,900

Gain if the price reaches our value

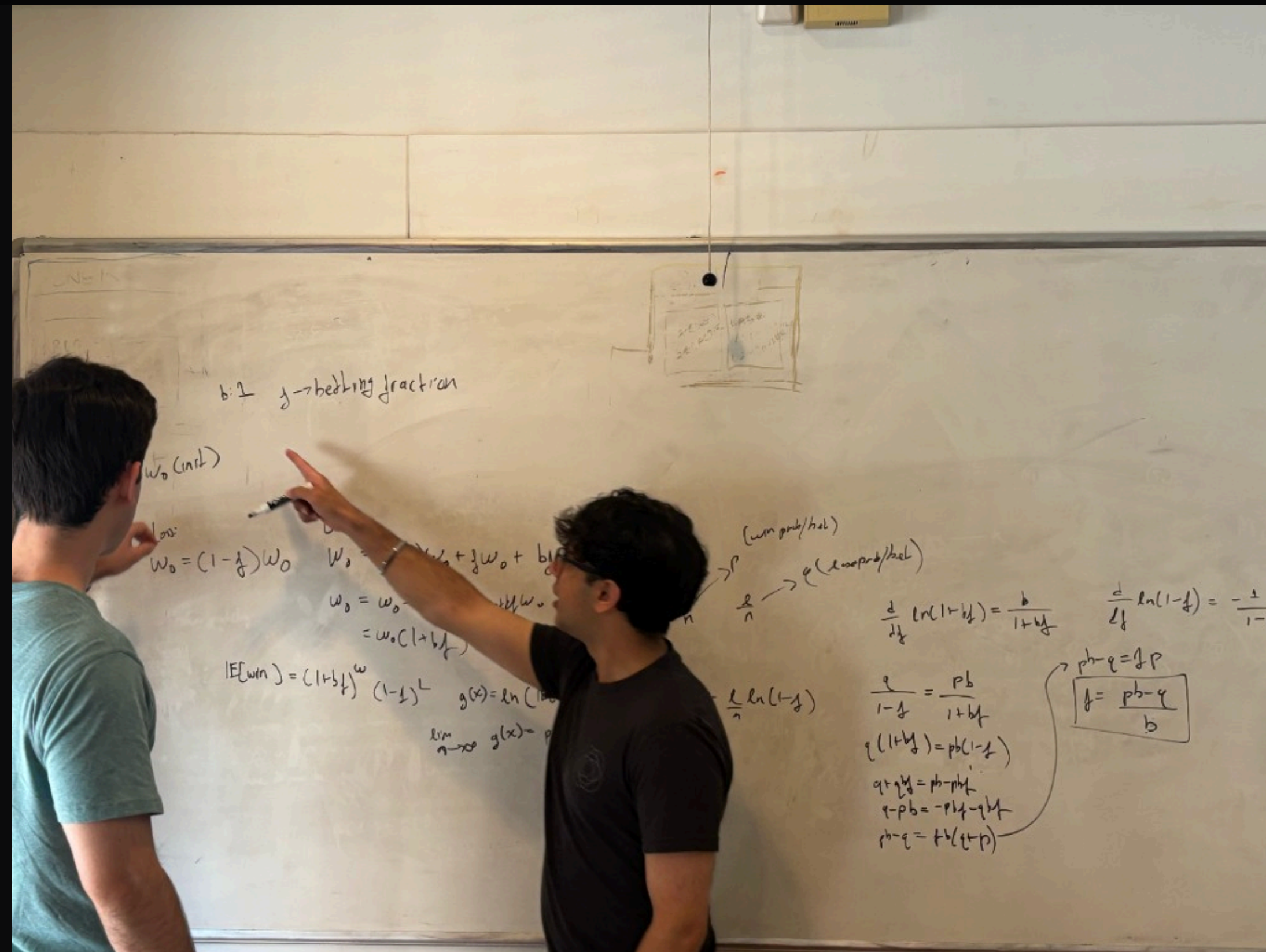
+\$508

Loss if our value is one miss too high

-\$4,194

cash \$100,000 · stake = slider x 0.28 x cash, capped at \$20,000

The stake is not a heuristic. It is the fraction that maximizes log wealth, then cut in half because the edge is



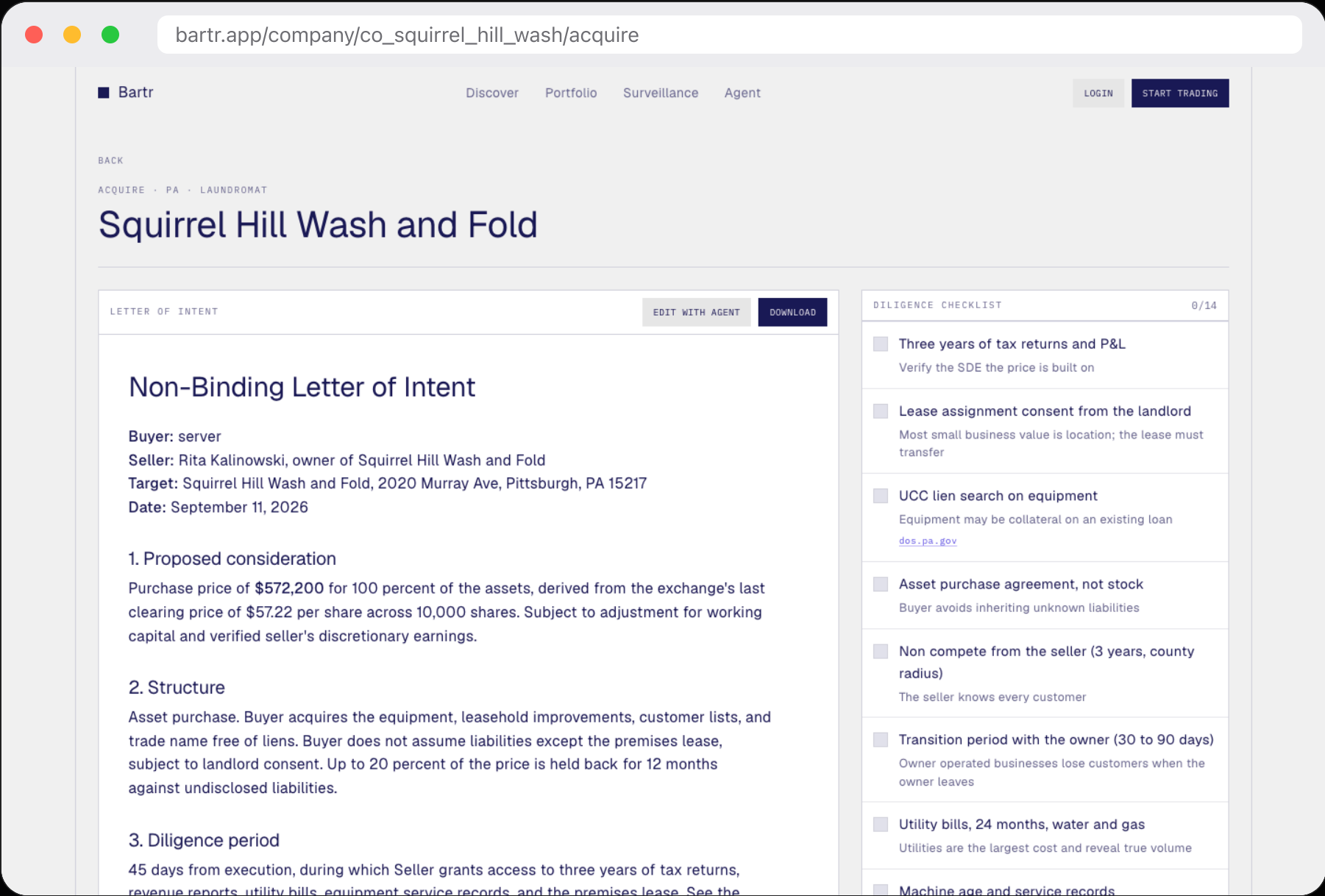
From 20 shares to the whole company.

Bartr drafts two things for you:

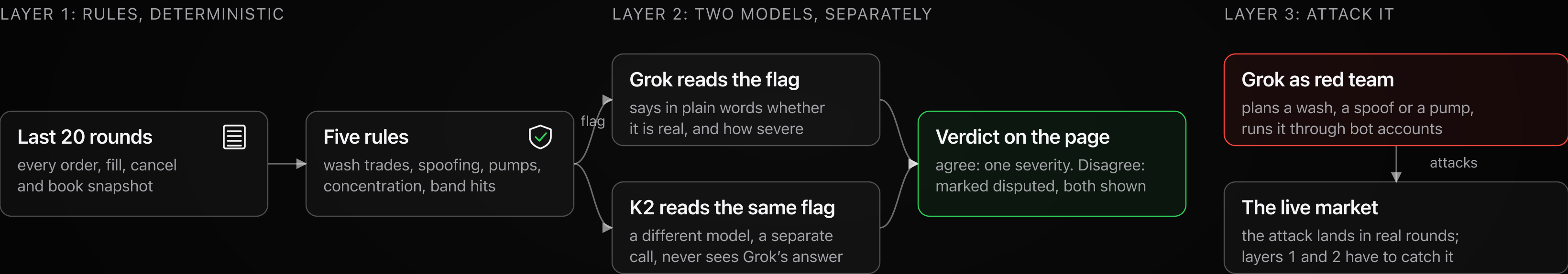
A letter of intent priced at the last traded price.

A diligence checklist for a Pittsburgh business: city registration, Allegheny County health permit, PA bulk sale clearance, lease, liens, tax returns.

Each government item links to the official page.



Rules catch it. Two models judge it. Grok tries to beat it.



● **Flagged** wash trading, high

Rule: wash_a and wash_b have traded 9 times and only with each other. The volume looks real but neither one's stake ever changed.

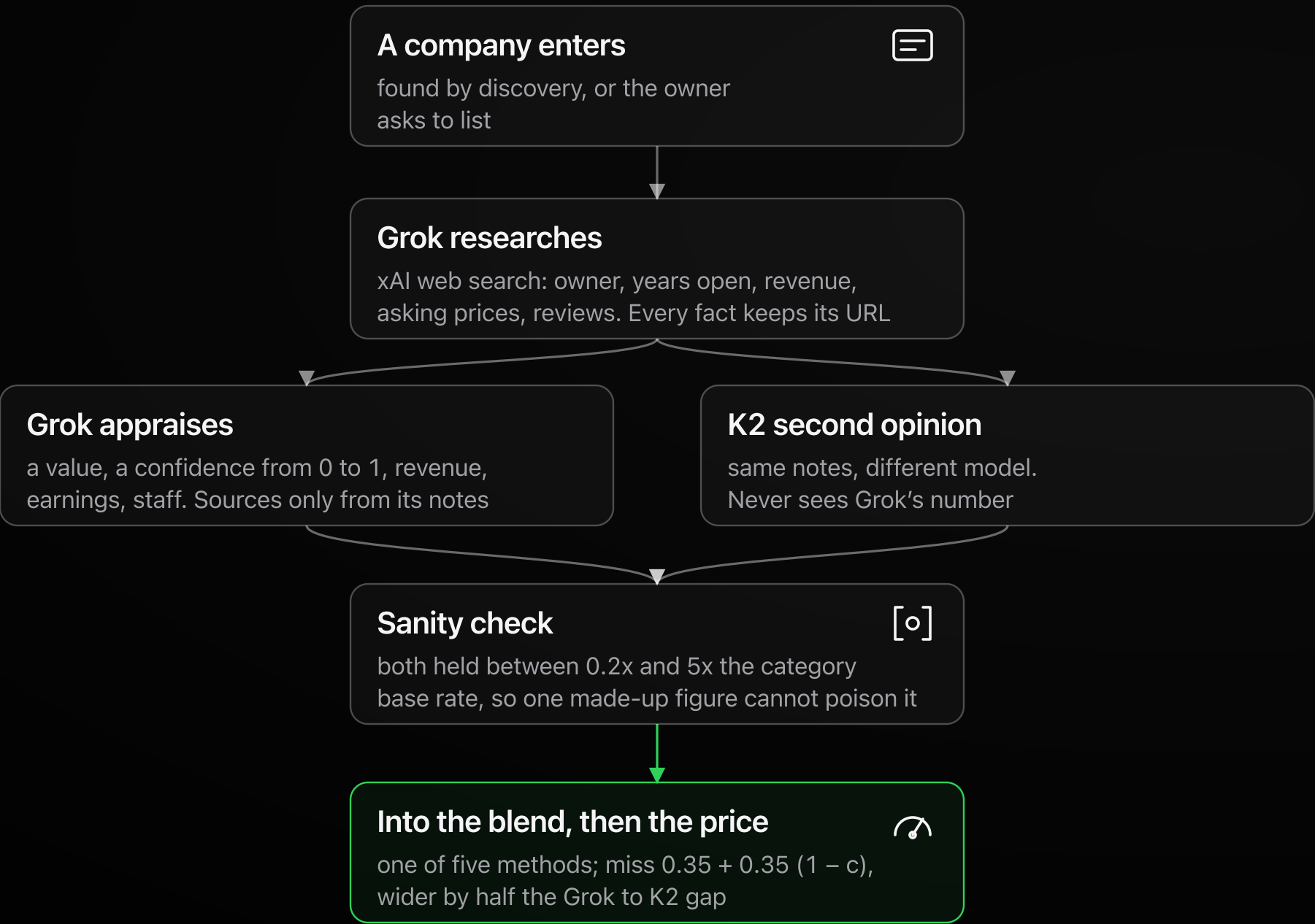
Grok high · K2 high · agree

Grok (example): "Alternating sides at the clearing price with no other counterparties. No economic purpose; treat as manipulation." K2 reached the same severity without seeing it.

What happens next

The flag and both reviews go on the admin dashboard and into the audit log. Two rounds in a row at the price band halt a market. Every red team attempt is logged with whether it was caught.

Every company that enters the exchange is appraised twice: Grok researches it and names a value, K2 names one without seeing Grok's, and both go into the price.



Everywhere else Grok runs

Reading the web. Builds a profile from Querit pages. A number is kept only if the exact sentence it came from is on the page, and financials must be in dollars with a date.

Security. Every rule flag goes to Grok and to K2 in separate calls. As red team, Grok plans an attack and runs it through bot accounts.

Chat. An agent with tools: search, open a company, read the book, place an order, get suggestions. The same agent answers over iMessage. Also an owner persona per company, a market narrator, and a market health memo.

Portfolio and acquire. Reads your profile into limits, writes the reason under each suggestion, drafts the letter of intent and the diligence checklist.

When Grok is down. Every call has a plain code fallback. Calls through the shared wrapper are cached and listed on /readiness.

How it runs

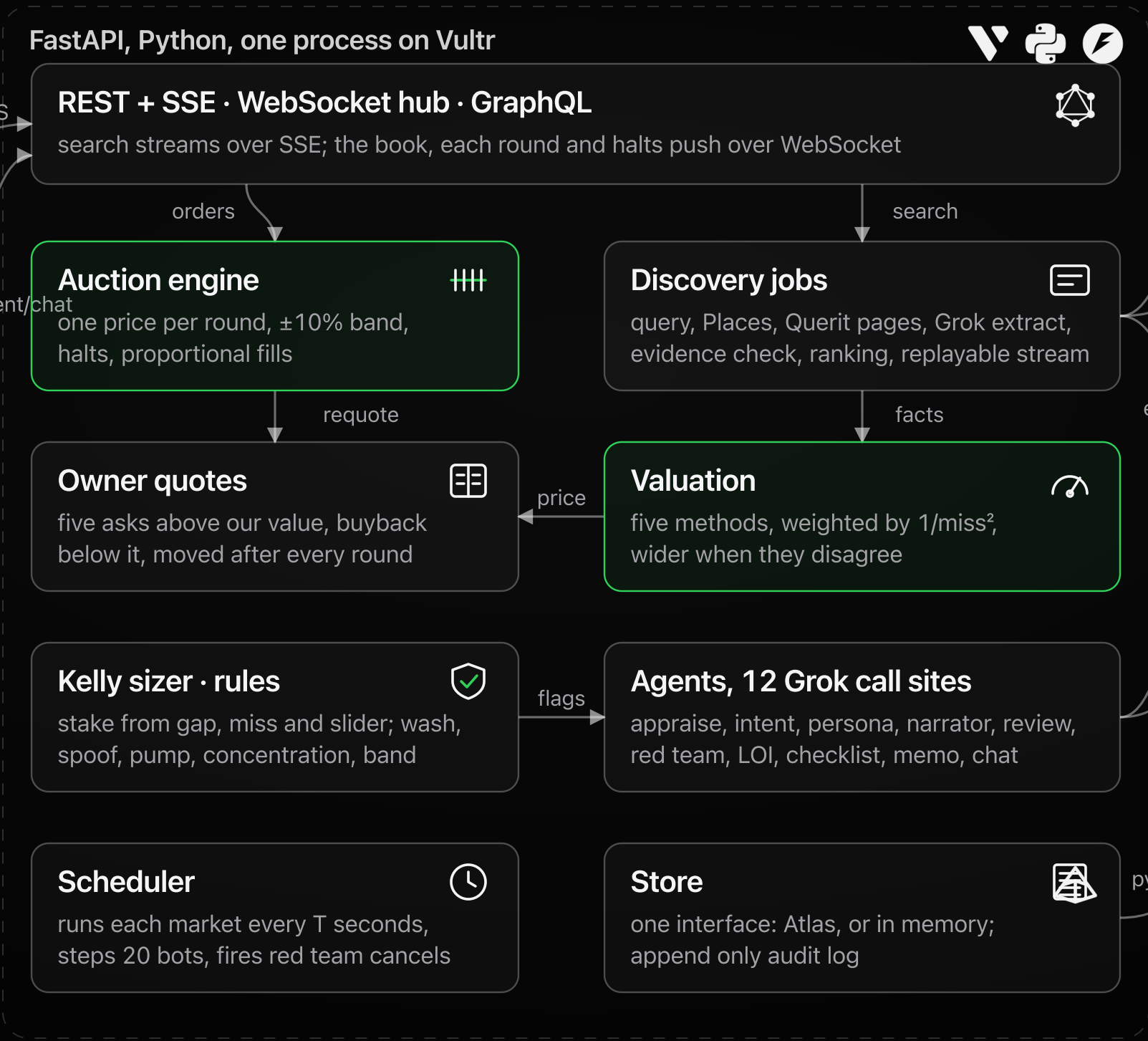
103 tests pass offline; every model call has a fallback

CLIENTS

Web app
Next.js 16 on Vercel. Discover, company, portfolio, acquire, surveillance, chat

iMessage
Photon bridge in Bun; texts go to the same chat agent

API



DATA AND MODELS

Google Places
businesses, ratings, review counts

Querit
reads the pages about each business

xAI · Grok
grok-4.6 with web search for research; a fast model for chat and reviews

IFM · K2 Horizon
second opinion on every appraisal and every flag

MongoDB Atlas
companies, markets, orders, rounds, trades, audit; migrations

Bartr.

The laundromat on Murray Avenue has a price. Buy 20 shares and watch the next round clear.

Grok
xAI

IFM K2
Horizon

Querit
web search

 **MongoDB Atlas**
database

 **Auth0**
login

 **Vultr**
API host

 **Vercel**
web host

 **Cursor**
editor

Q&A preparation

Stop the public presentation on the close slide. Press End. Use this appendix only for questions.

Answer first Give the conclusion in one sentence. Explain the mechanism only if the judge wants more.

Name the limit Say what works today, what is simulated, and what requires a production partner.

Do not bluff If you do not know, state the assumption and describe the test that would answer it.

Hand off cleanly Nico: discovery and valuation. Vir: data and infrastructure. Aditya: auction, security, and regulation. Zhiyuan: product, frontend, and demo.

Is there a real market behind the demo?

Q Why does this need to exist now?	Ownership is about to move. The US has 36.2 million small businesses. Project Equity estimates 2.9 million businesses have owners aged 55 or older, supporting 32.1 million jobs and \$6.5 trillion in annual revenue. Better web data and models now make the hidden supply searchable before an owner hires a broker.
Q Is 36.2 million really your TAM?	No. It is the scale of the information gap, not a revenue forecast. Our first reachable market is profitable employer businesses in repeatable local categories, beginning with Pittsburgh laundromats, car washes, service companies, and small manufacturers.
Q What proves buyer demand?	Existing marketplaces already attract buyers. BizBuySell reports more than 4 million visits each month, while only 2,117 transactions were reported in Q2 2026. Its own report says demand exceeds the supply of acquisition-ready businesses.
Q What is the narrow first wedge?	Acquisition search in one dense city. Help serious buyers find and compare businesses that are not advertised for sale. Owner-approved fractional trading comes later through a regulated partner.

Who wants this, and why would owners cooperate?

Q Who is the first paying customer?	The first user is a serious buyer or acquirer. They pay for broader discovery, evidence-backed comparison, and a qualified path to contact. The owner becomes a customer only after real interest exists.
Q Why would an owner sell only 30%?	Partial liquidity lets an owner take cash out without giving up control. Thirty percent is a demo parameter, not a universal answer. The real percentage would depend on owner goals, governance rights, tax structure, and offering rules.
Q Can you publish a business without permission?	We can organize public facts, but we should not imply endorsement. An unclaimed page must say that Bartr produced the estimate, provide sources, accept corrections and opt-outs, and prohibit trading until the owner verifies and accepts the listing.
Q Are you impersonating the owner with the owner persona?	No production system should do that without authorization. The current persona is a demo simulation grounded in the company file. A real product would label it as a Bartr assistant until the owner claims the page and authorizes responses.

Why does an existing platform not already solve this?

Q Why not just use Google Maps?	Maps finds storefronts, not investable businesses. It does not assemble ownership evidence, operating estimates, valuation ranges, comparable prices, an offer workflow, or diligence requirements.
Q Why not BizBuySell?	BizBuySell starts after an owner decides to sell. Bartr starts with the much larger set of businesses that are not listed, prices uncertainty, and lets qualified buyer interest begin the owner conversation.
Q Why not StartEngine or another crowdfunding portal?	Crowdfunding primarily helps an issuer raise money. Bartr begins with acquisition discovery and valuation. Reg CF securities also generally carry a one-year resale restriction, and a funding portal cannot give investment recommendations or operate a secondary exchange.
Q What stops an incumbent from copying this?	The code alone is not a moat. Defensibility would come from verified owner relationships, a growing record of valuation errors and completed transactions, category-specific evidence, and regulated distribution partnerships.

How can you price a private business with incomplete data?

Q How can you value a business without its books?

We show an estimate and a wide range, not a certified valuation. Before owner verification, the model uses public evidence, category multiples, observable proxies, and independent model opinions. Missing evidence increases uncertainty. Trading stays closed until the owner accepts the listing.

Q Have you proven that the estimates are accurate?

Not yet at production scale. We built a holdout calibration path that hides known listing prices and measures each estimator's error, but we do not yet have enough completed observations to claim empirical calibration. That is an explicit next validation milestone.

Q What stops Grok from hallucinating revenue?

A number only enters verified evidence when the exact supporting sentence, source URL, unit, and date are present. Grok is one estimator, not the final answer. K2 gives a blind second opinion, deterministic methods remain in the blend, and disagreement widens the range.

Q Does market price create a circular valuation?

It can if thin trades receive too much weight. The model begins with an evidence-based prior and weights market updates by volume. Production should also cap the influence of concentrated or flagged volume and keep the model value separate from the traded price.

Q Where does the Kelly fraction come from?

From the log-wealth derivation on the board, not from a blog post. Wealth after a win is $W(1+bf)$, after a loss $W(1-f)$. Maximizing $E[\ln W]$ gives $f^* = (pb-q)/b$. We use half of that, capped at 20 percent, because the edge is estimated. Scroll the paper on the next slide.



1

Kelly Criterion Derivation:

$w_0 \rightarrow$ init capital $b \rightarrow$ percent return (excluding init)
(return if we win)

$f \rightarrow$ optimal fraction to bet

$$\begin{aligned}
 \text{If won: } w_1 &= (1-f)w_0 + fw_0 + fbw_0 & \text{If lost: } \underline{w_1} &= \underline{(1-f)w_0} \\
 &= w_0 - \cancel{fw_0} + \cancel{fw_0} + fbw_0 \\
 &= \underline{w_0(1-f)}
 \end{aligned}$$

$w = \# \text{ wins}$ $L = \# \text{ losses}$

What happens when a tiny private market meets reality?

Q Where does liquidity come from?

In the demo, the owner posts a finite ask ladder and a finite buyback bid. Bots make the demonstration active. Production cannot promise liquidity. It must show available size, funding, quote age, and transfer restrictions clearly.

Q What if the owner's buyback money runs out?

The bid disappears when its funded quantity is exhausted. A production bid should be limited by prefunded or verified capital. We should call it an owner repurchase order, never a guaranteed exit.

Q Why a batch auction instead of normal continuous trading?

A batch gives every participant in the round one clearing price. Speed does not decide priority, aggressive orders receive price improvement, and the engine chooses the price that executes the most volume. That fits a thin local market better than a race for the first quote.

Q Can a few users manipulate the price?

Yes, thin markets remain vulnerable. Bartr reduces the risk with identity controls, self-trade prevention, position limits, a 10% band, proportional fills, concentration rules, halts, and human review. Surveillance reduces risk. It does not eliminate it.

Can Bartr legally sell and trade these shares?

Q Are these shares securities?	Yes. A live-money product cannot treat 10,000 shares as a software abstraction. Every offer and sale needs registration or an exemption, and matching secondary orders generally requires regulated market infrastructure.
Q What is the realistic launch structure?	Bartr supplies discovery, valuation, auction technology, and surveillance. A registered intermediary conducts the offering. A broker-dealer-operated ATS executes eligible secondary trades. A transfer agent and custodian maintain ownership and assets, with KYC and AML controls.
Q Does Regulation Crowdfunding solve everything?	No. Reg CF can let an issuer raise up to \$5 million through a registered intermediary, but the securities generally cannot be resold for one year. It does not automatically cover an owner's personal 30% cash-out. Reg A or another exemption may fit that structure better.
Q Can Bartr give Kelly-sized recommendations?	Personalized recommendations about specific securities may be investment advice. A funding portal is prohibited from making recommendations. Production would need a broker-dealer or adviser structure, or would limit Bartr to user-directed, non-personalized analytics.

What does an investor actually own?

Q What legal rights come with one share?

The demo models economic units, not a finished security. A real offering must define voting rights, distributions, information rights, transfer restrictions, dilution, repurchase terms, and priority relative to debt and the owner's retained shares.

Q How does an investor receive returns?

That depends on the security design. Returns could come from declared distributions, a later repurchase, or a sale. Bartr should not imply automatic dividends or price appreciation before the operating agreement and disclosure documents define them.

Q Who maintains the cap table and tax records?

A production transfer agent should maintain the official holder record. A tax and fund-administration partner would handle statements and any K-1 or 1099 obligations. MongoDB is the app ledger today, not the final legal ownership system.

Q Does owning 20 shares make acquisition easier?

It creates information and a relationship, not a right to buy the company. The owner still chooses whether to negotiate. The LOI is nonbinding, the final price comes from diligence and negotiation, and a licensed M&A process may be required.

How would Bartr become a business?

Q How do you make money?	The first hypothesis is paid acquisition intelligence for serious buyers. Later revenue could include owner verification and valuation services plus regulated transaction fees shared with licensed partners. Bartr should not earn a trading spread because it does not trade against users.
Q How do you solve the marketplace cold start?	Discovery creates useful inventory before owners join, but trading stays closed. Qualified buyer interest generates a specific owner outreach. Accepted owners create the first verified supply. Density in one city and a few categories matters more than national coverage.
Q How expensive will owner acquisition be?	We do not know yet. The key experiment is the funnel from qualified buyer interest to owner response, verified financials, and accepted listing. Generic cold outreach would be expensive and damage trust.
Q What metrics would prove the business?	Owner response and verification rates come first. Then measure valuation error, time to first qualified offer, cost per verified business, repeat buyer usage, and real liquidity without simulated traders.

What is real, and what was built for the demo?

Q What actually works end to end?	The FastAPI service runs discovery jobs, valuation, order books, batch clearing, belief updates, Kelly sizing, acquisition documents, rules, model reviews, WebSockets, and audit events. The Next.js app uses those routes when configured.
Q What is simulated or preloaded?	The demo includes seeded companies, cached research, 20 bot traders, planted manipulation, \$100,000 demo balances, and X-Demo-User identity. No real money or legal securities settle.
Q Can the current architecture scale?	Not as a production exchange. It deliberately runs one API worker with an in-process scheduler and WebSocket hub. Production needs durable jobs, atomic settlement, separate market workers, reconciliation, disaster recovery, and load testing.
Q What happens when Grok, K2, or Querit fails?	Interactive paths have deterministic fallbacks and cached outputs. Search preserves partial results. Research calls can return unavailable rather than invent evidence. The readiness endpoint exposes provider status and recent calls.

How much protection is real today?

Q Can someone create many accounts and manipulate a market?	Yes in the demo because identity is intentionally lightweight. Production requires broker-level customer identification, beneficial-owner checks, sanctions screening, linked-account detection, funding controls, and device or network signals.
Q Why trust AI to police manipulation?	AI does not enforce trades. Deterministic rules create the flags. Grok and K2 independently explain and grade the evidence. Disagreement is shown to a human rather than averaged away.
Q Is the audit log truly immutable?	It is append-only through the application today, not cryptographically immutable. Production needs restricted write roles, retention controls, tamper-evident hashes or WORM storage, independent backups, and reconciliation with broker and transfer-agent records.
Q How do you protect private financial data?	The current demo mostly uses public and seeded data. Production needs field-level access control, encryption, consent records, redaction, retention limits, vendor review, incident response, and strict separation between public estimates and owner-provided documents.

What could make the entire thesis wrong?

Q What is the biggest unproven assumption?	That owners will exchange verified financial data and partial equity for qualified demand. Search and valuation are technically feasible. Owner trust and conversion determine whether a marketplace exists.
Q What results would make you stop?	We would reconsider the exchange model if owners reject partial liquidity, verified valuations remain too wide to support decisions, buyers do not return, or regulatory cost overwhelms the economics of a typical \$350,000 transaction.
Q Could a public estimate harm a local business?	Yes. That is why estimates need ranges, evidence, clear Bartr attribution, correction and opt-out paths, and no trading before owner consent. Sensitive inferred financials may need to remain private until verification.
Q What are the next concrete proofs?	Interview owners and acquisition buyers. Validate estimates on held-out real transactions. Secure a broker, ATS, and securities-counsel design. Run a clearly labeled paper market. Convert the first five owner-verified Pittsburgh businesses.